4. Deployment of Machine learning Models

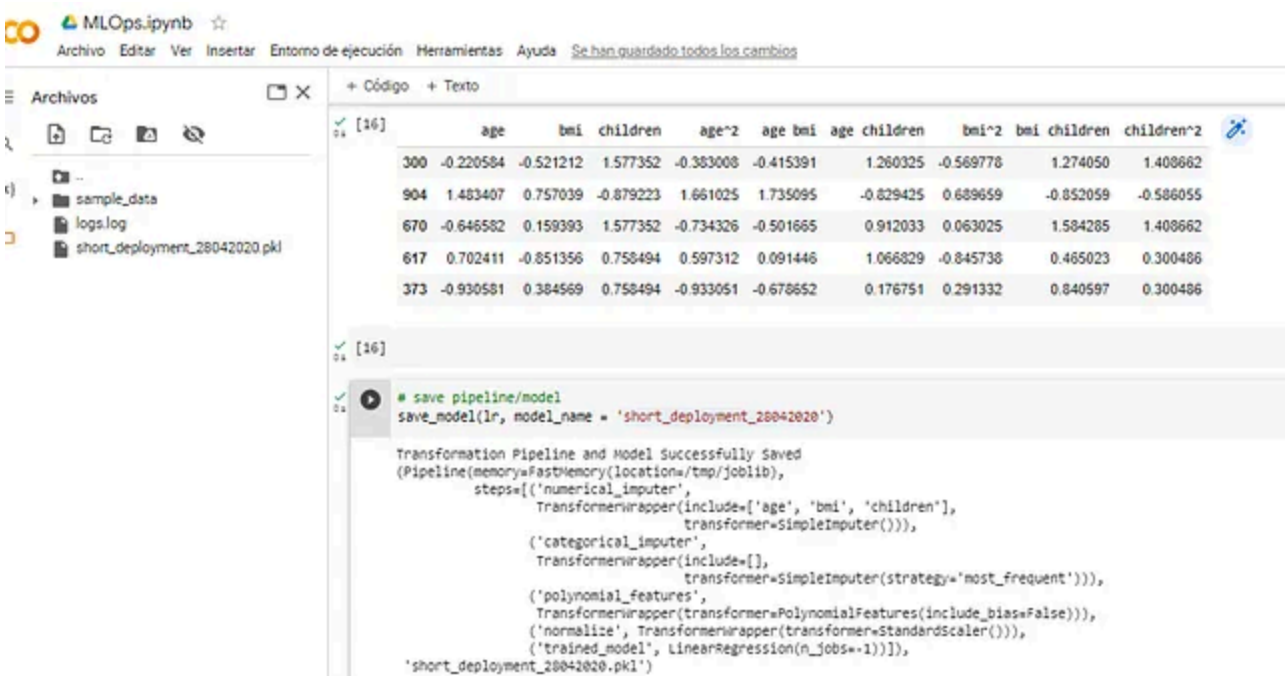
Here we will create a model with pycaret in Google Collab, and we will deploy it in a Web App with Flask and publish a docker image and Web Application in Azure.

1. Train and save a model with Google Collab.

```
# save pipeline/model
save_model(lr, model_name = 'short_deployment_28042020')
```

https://github.com/armandoordonez/mlops/blob/main/MLOps.ipynb

2. Download the trained model (short_deployment_28042020)



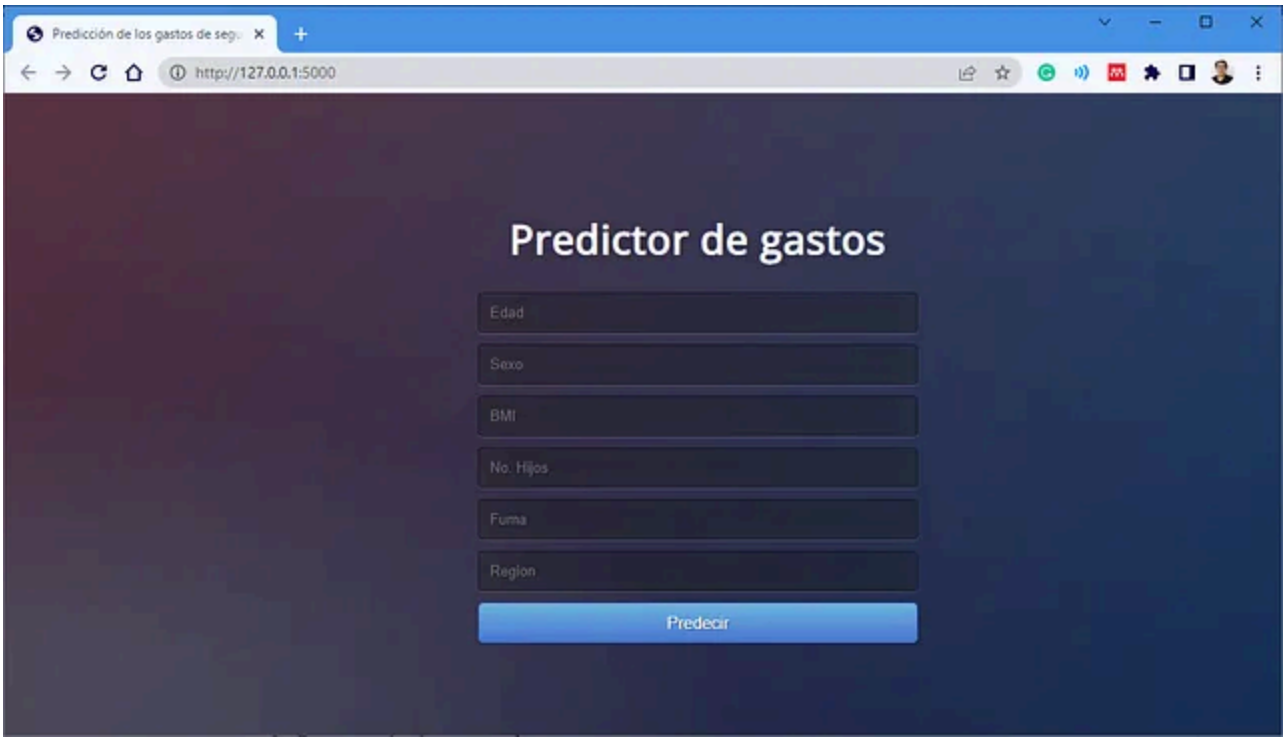
	age	bmi	children	age^2	age bmi	age children	bmi^2	bmi children	children^2
300	-0.220584	-0.521212	1.577352	-0.383008	-0.415391	1.260325	-0.569778	1.274050	1.408662
904	1.483407	0.757039	-0.879223	1.661025	1.735095	-0.829425	0.689659	-0.852059	-0.586055
670	-0.646582	0.159393	1.577352	-0.734326	-0.501665	0.912033	0.063025	1.584285	1.408662
617	0.702411	-0.851356	0.758494	0.597312	0.091446	1.066829	-0.845738	0.465023	0.300486
373	-0.930581	0.384569	0.758494	-0.933051	-0.678652	0.176751	0.291332	0.840597	0.300486

3. Create a Flask app locally (For example in Visual Studio Code)

```
from flask import Flask,request, url_for, redirect, render_template, jsonify
from pycaret.regression import *
import pandas as pd
import pickle
import numpy as np

app = Flask(__name__)
```

```
model = load_model('deployment_28042020')
cols = ['age', 'sex', 'bmi', 'children', 'smoker', 'region']

@app.route('/')
def home():
    return render_template("home.html")

@app.route('/predict',methods=['POST'])
def predict():
    int_features = [x for x in request.form.values()]
    final = np.array(int_features)
    data_unseen = pd.DataFrame([final], columns = cols)
    prediction = predict_model(model, data=data_unseen, round = 0)
    # prediction = 1540000
    #print("***************",prediction)
    prediction = int(prediction.prediction_label[0])
    return render_template('home.html',pred='Los gastos esperados son
{}'.format(prediction))

@app.route('/predict_api',methods=['POST'])
def predict_api():
    data = request.get_json(force=True)
    data_unseen = pd.DataFrame([data])
    prediction = predict_model(model, data=data_unseen)
    #prediction = 1540000
    output = prediction.Label[0]
    return jsonify(output)

if __name__ == '__main__':
    app.run(debug=True)
```

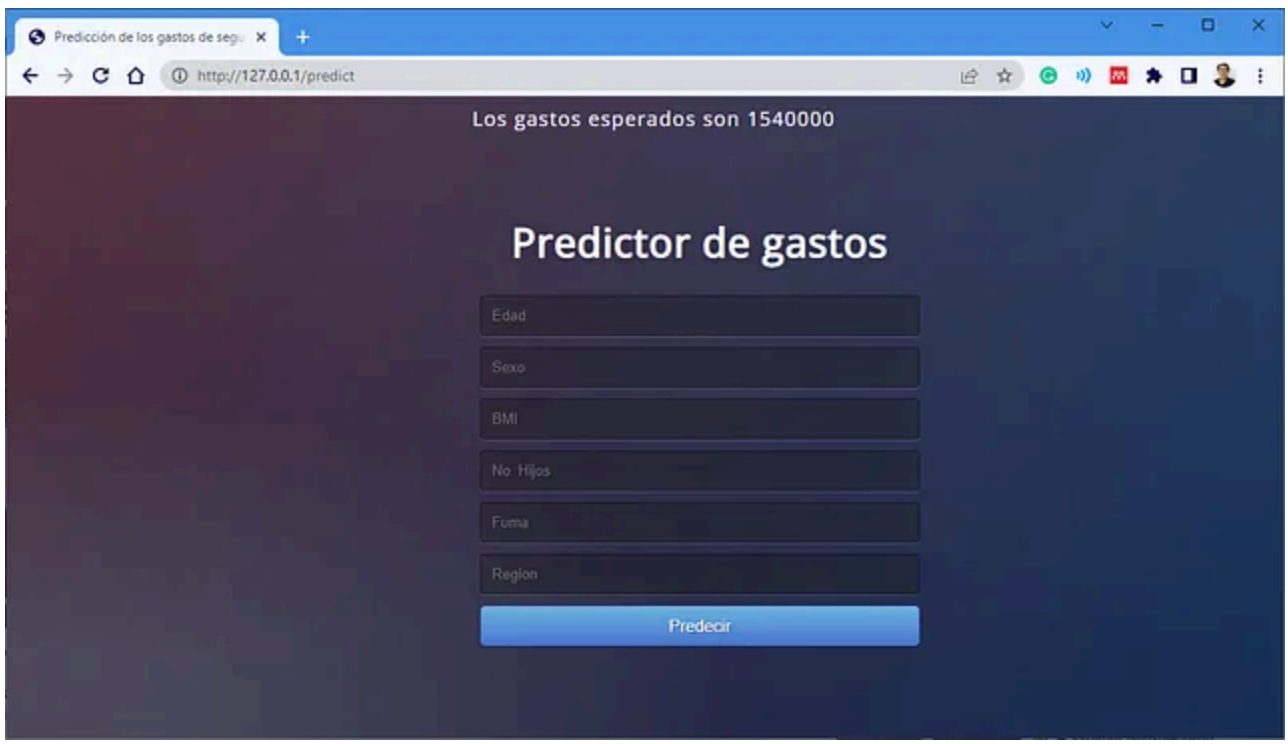
4. Run the application locally: http://127.0.0.1:5000/



5. **Modify the code to make it light:** just for academic purposes we will make a light version of the flask server, this will ease or deployment process.

```
from flask import Flask,request, url_for, redirect, render_template, jsonify
# from pycaret.regression import *
# import pandas as pd
# import pickle
# import numpy as np
import os

app = Flask(__name__)

# model = load_model('deployment_28042020')
cols = ['age', 'sex', 'bmi', 'children', 'smoker', 'region']

@app.route('/')
def home():
    return render_template("home.html")

@app.route('/predict',methods=['POST', 'GET'])
def predict():
    #int_features = [x for x in request.form.values()]
    #final = np.array(int_features)
    #data_unseen = pd.DataFrame([final], columns = cols)
    # prediction = predict_model(model, data=data_unseen, round = 0)
```

```
    prediction = 1540000
    #print("***************",prediction)
    # prediction = int(prediction.prediction_label[0])
    return render_template('home.html',pred='Los gastos esperados son
{}'.format(prediction))

@app.route('/predict_api',methods=['POST'])
def predict_api():
    #data = request.get_json(force=True)
    #data_unseen = pd.DataFrame([data])
    #prediction = predict_model(model, data=data_unseen)
    prediction = 1540000
    #output = prediction.Label[0]
    return jsonify(prediction)

if __name__ == '__main__':
    port = int(os.environ.get('PORT', 80))
    app.run(debug=True, host='0.0.0.0', port=port)
```



6. Create a file called Dockerfile (without extensions)

```
# start by pulling the python image
FROM python:3.8-alpine

RUN pip install --upgrade pip
```

```
RUN pip install virtualenv

ENV VIRTUAL_ENV=/venv
RUN virtualenv venv -p python3

ENV PATH="VIRTUAL_ENV/bin:$PATH"

WORKDIR /app
ADD . /app

# install dependencies
RUN pip install -r requirements.txt

# expose port
EXPOSE 80

# configure the container to run in an executed manner
ENTRYPOINT [ "python" ]

CMD ["app_light.py" ]

# run application
# CMD ["python", "app_light.py"]
```

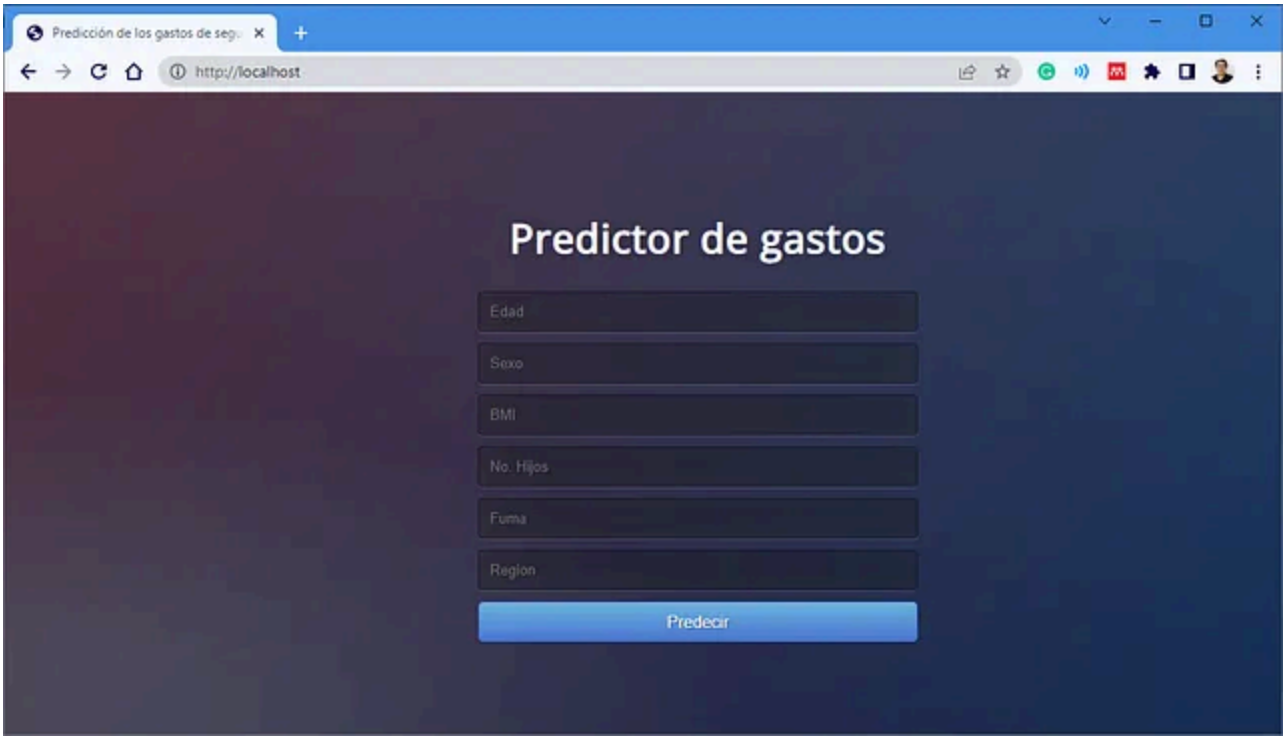
8. Test the container locally (You need a docker installation in your own machine)

```
cd Dropbox\Courses\MLOps\deployment-heroku-master\mlops

docker build -t mlopsregistro130423.azurecr.io/gastos:latest .
```

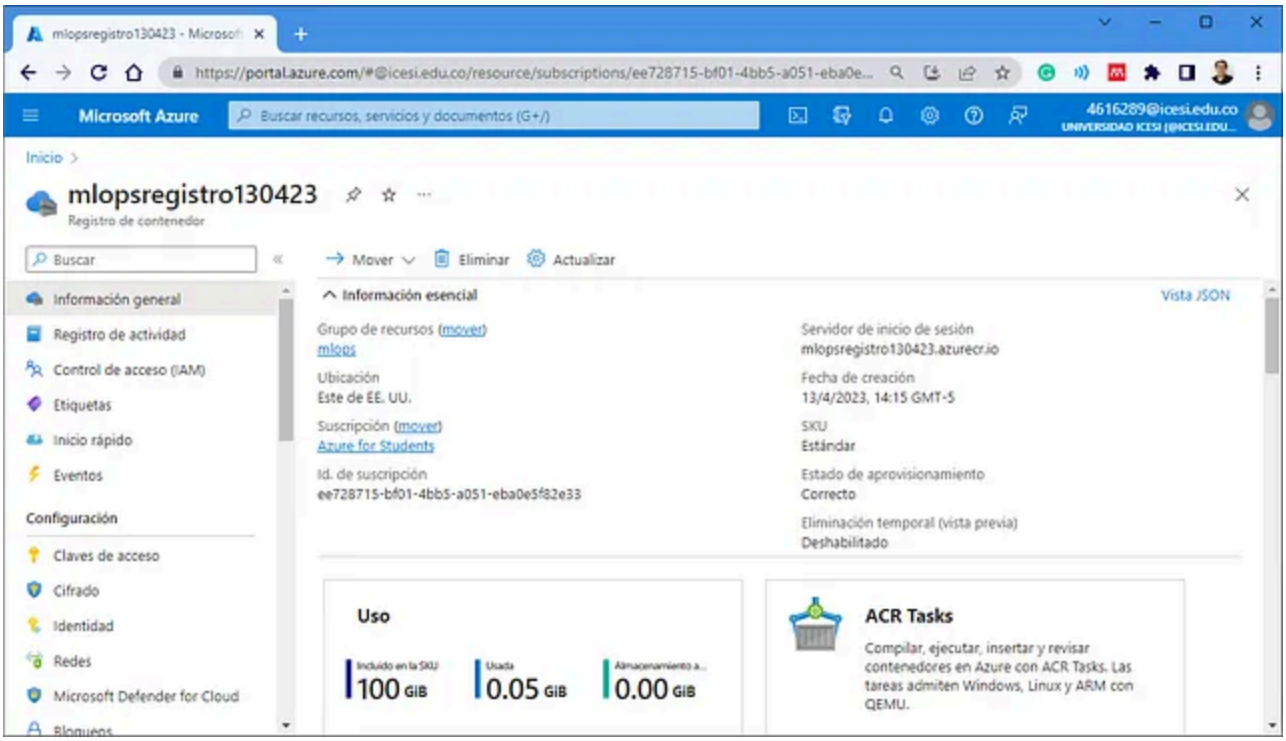
```
C:\Windows\system32\cmd.exe
r.io/gastos
Unable to find image 'mlopsregistro130423.azurecr.io/gastos:latest' locally
docker: Error response from daemon: manifest for mlopsregistro130423.azurecr.io/gastos:latest not found: manifest unknow
n: manifest tagged by "latest" is not found.
See 'docker run --help'.

C:\Users\Personal\Dropbox\Courses\MLOps\deployment-heroku-master\mlops>docker build -t mlopsregistro130423.azurecr.io/ga
stos:latest .
[+] Building 12.9s (12/12) FINISHED
 => [internal] load build definition from Dockerfile                                                    0.0s
 => => transferring dockerfile: 32B                                                                     0.0s
 => [internal] load .dockerignore                                                                       0.0s
 => => transferring context: 2B                                                                         0.0s
 => [internal] load metadata for docker.io/library/python:3.8-alpine                                    1.4s
 => [1/7] FROM docker.io/library/python:3.8-alpine@sha256:cd2afe99fc96d851a30ee3956a3e12c78c0fe766ab4afd47f96e5ca  0.0s
 => [internal] load build context                                                                       0.2s
 => => transferring context: 177.67kB                                                                   0.1s
 => CACHED [2/7] RUN pip install --upgrade pip                                                          0.0s
 => CACHED [3/7] RUN pip install virtualenv                                                             0.0s
 => CACHED [4/7] RUN virtualenv venv -p python3                                                         0.0s
 => CACHED [5/7] WORKDIR /app                                                                           0.0s
 => [6/7] ADD . /app                                                                                    0.3s
 => [7/7] RUN pip install -r requirements.txt                                                          10.6s
 => exporting to image                                                                                  0.3s
 => => exporting layers                                                                                 0.3s
 => => writing image sha256:9fb4a98634df9319a84c6763ac144683de0f503f11b5f164d9d351a8c003be98             0.0s
 => => naming to mlopsregistro130423.azurecr.io/gastos:latest                                           0.0s

C:\Users\Personal\Dropbox\Courses\MLOps\deployment-heroku-master\mlops>
```

```
docker run -d -p 80:80 mlopsregistro130423.azurecr.io/gastos
```

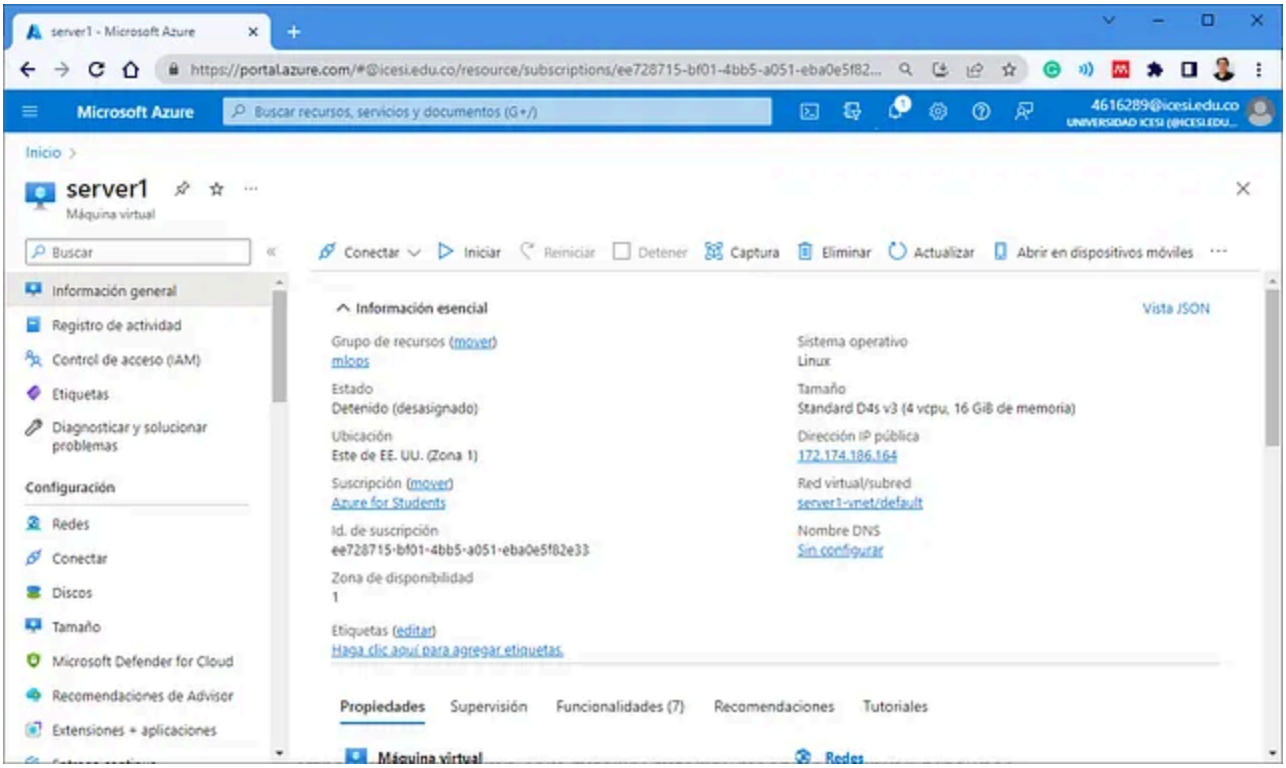
```
C:\Windows\system32\cmd.exe

C:\Users\Personal\Dropbox\Courses\MLOps\deployment-heroku-master\mlops>docker run -d -p 80:80 mlopsregistro130423.azurec
r.io/gastos
897ea815604e68c401805b3f23c8a2a5d6eb0224570b1e528a73396020f6341b

C:\Users\Personal\Dropbox\Courses\MLOps\deployment-heroku-master\mlops>_
```



9. Create a docker repository in Azure

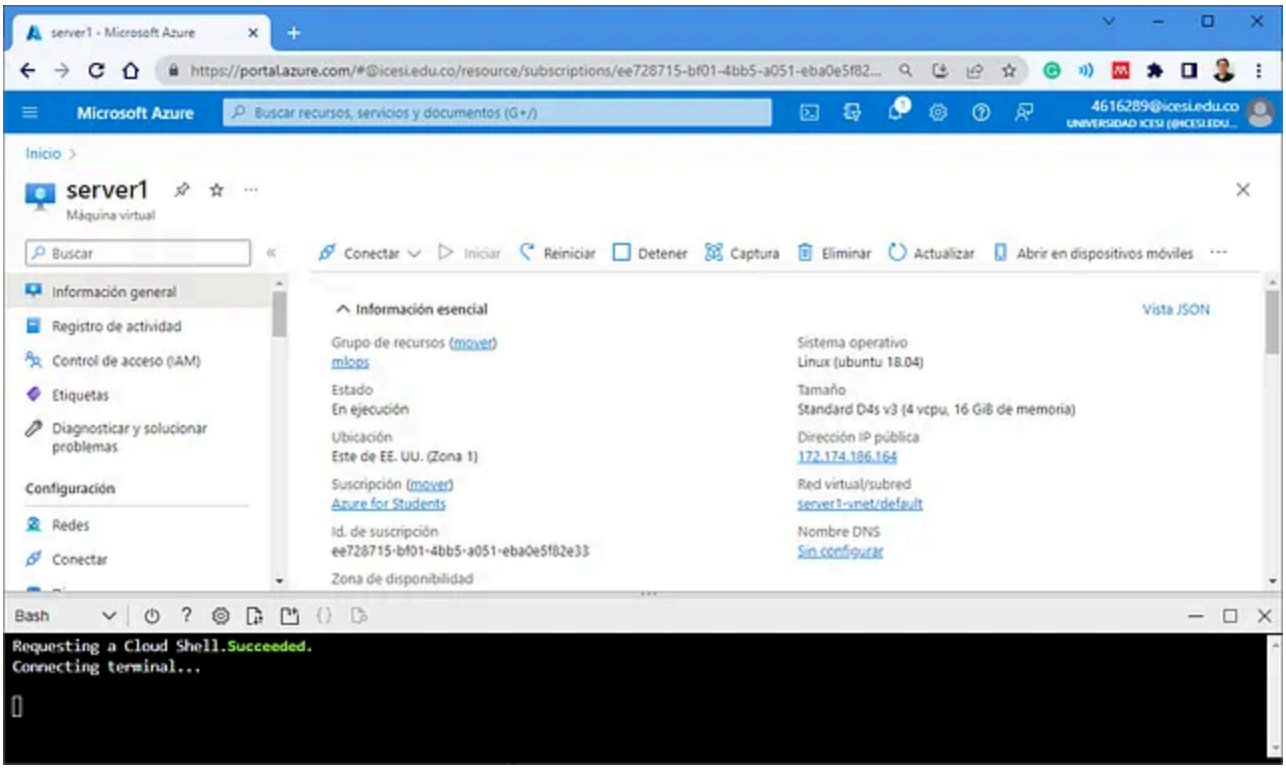


10. Create a virtual machine



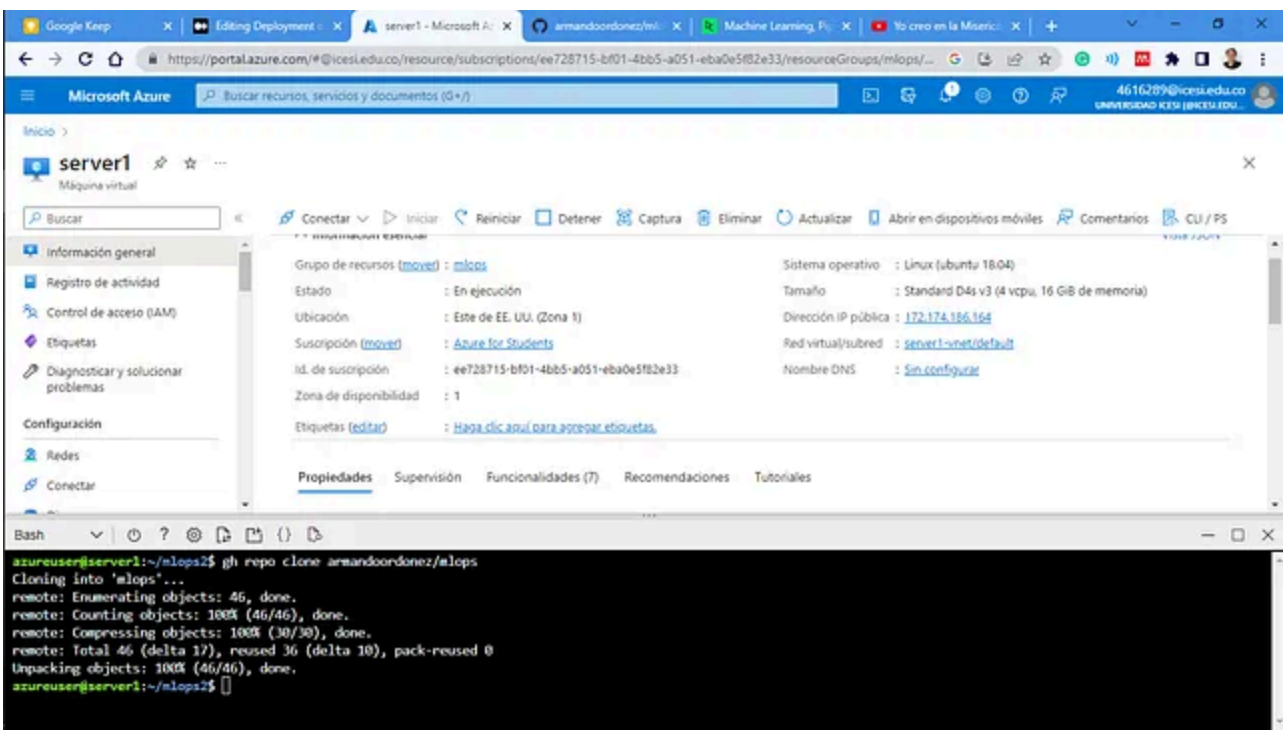
11. Connect to the machine

```
ssh -i server1_key.pem azureuser@172.174.186.164
```



12. Clone the code from github

```
gh repo clone armandoordonez/mlops
```

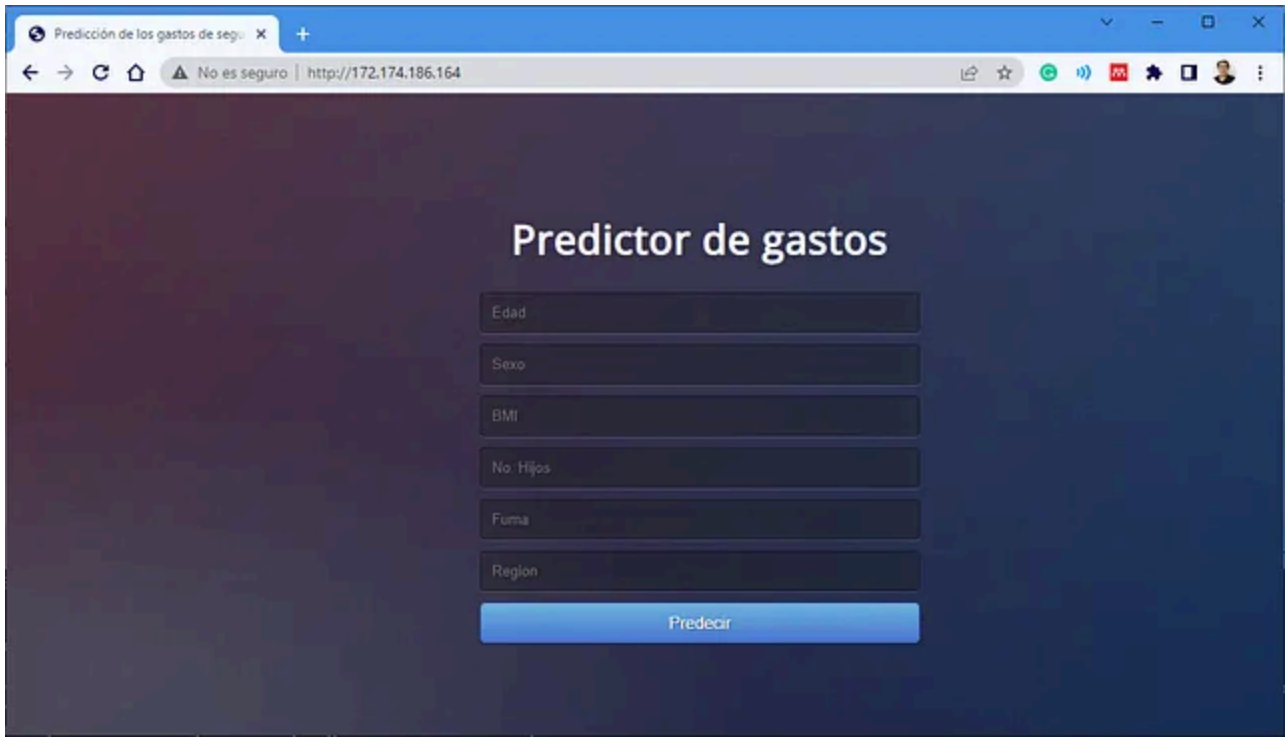


13. Create an image and a container in the new server using the previous commands.

```
azureuser@server1:~$ docker ps
CONTAINER ID   IMAGE      COMMAND   CREATED   STATUS    PORTS     NAMES
```

```
azureuser@server1:~$ docker image ls
REPOSITORY                                TAG       IMAGE ID       CREATED
SIZE
mlopsregistro130423.azurecr.io/costos   latest    bd2fbf26fc5b   3 days ago
162MB
ubuntu                                    latest    08d22c0ceb15   5 weeks ago
77.8MB
hello-world                               latest    feb5d9fea6a5   19 months ago
13.3kB
azureuser@server1:~$
```

14. Try the app on the virtual server



15. Create a Web App from docker

Crear aplicación web - Microsoft

https://portal.azure.com/#create/Microsoft.WebSite

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Inicio > Crear un recurso >

Crear aplicación web

Detalles del proyecto

Seleccione una suscripción para administrar los recursos implementados y los costos. Use los grupos de recursos como carpetas para organizar y administrar todos los recursos.

Suscripción * : Azure for Students

Grupo de recursos * : mlops

Crear nuevo

Detalles de instancia

Revisar y crear | < Anterior | Siguiente: Docker >

Bash

```
docker: Error response from daemon: manifest for mlopsregistro130423.azurecr.io/gastos:latest not found: manifest unknown: manifest tagged by
"latest" is not found.
See 'docker run --help'.
azureuser@server1:~$ docker run -d -p 80:80 mlopsregistro130423.azurecr.io/costos
4304c4358f32d3d0358d6d6f91d1bed41a60e046a4cf2710e3f16083aa4525e4
azureuser@server1:~$
```

Crear aplicación web - Microsoft

https://portal.azure.com/#create/Microsoft.WebSite

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Inicio > Crear un recurso >

Crear aplicación web

Datos básicos | **Docker** | Redes | Supervisión | Etiquetas | Revisar y crear

Extrae imágenes de contenedor de Azure Container Registry, Docker Hub o un repositorio de Docker privado. App Service implementará la aplicación en contenedores con sus dependencias preferidas en producción en cuestión de segundos.

Opciones: Contenedor único

Origen de imagen: Azure Container Registry

Opciones de Azure Container Registry

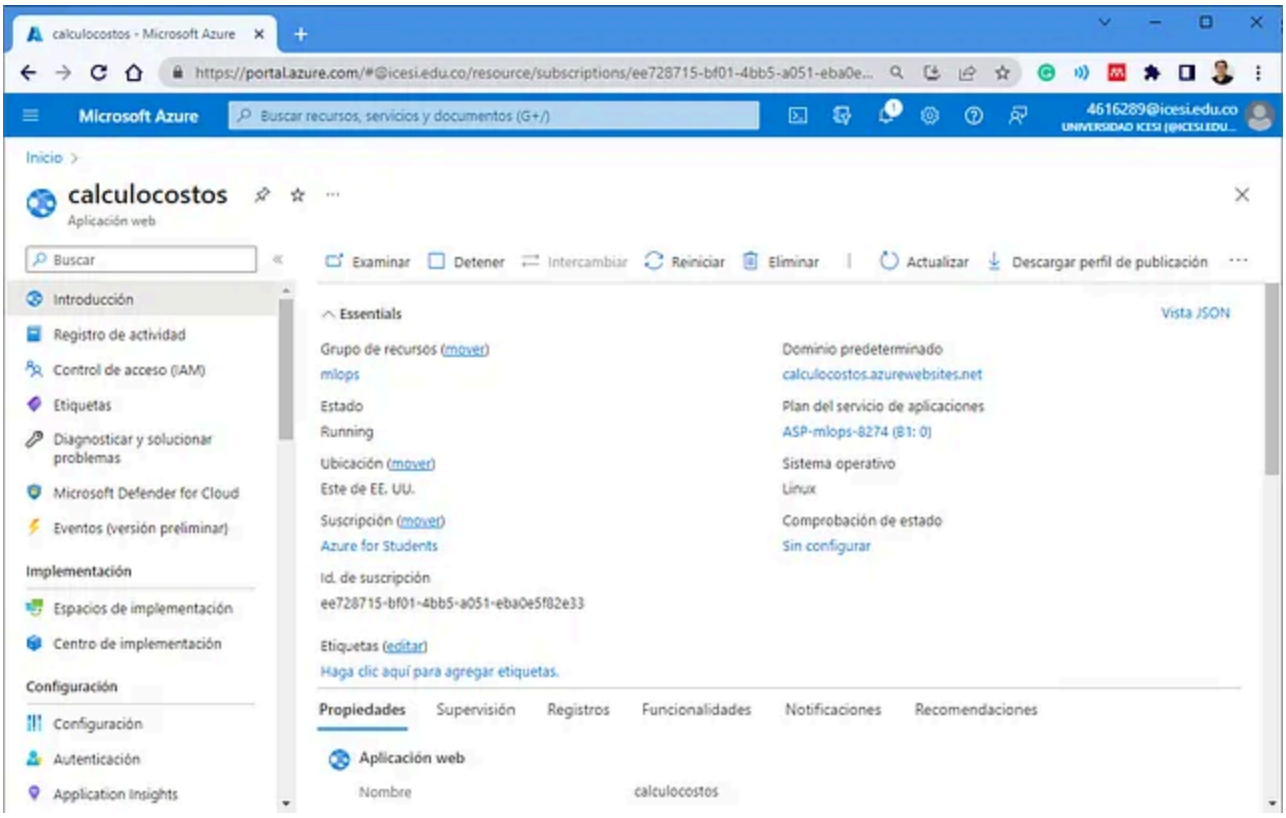
Registro * : mlopsregistro130423

Imagen * : costos

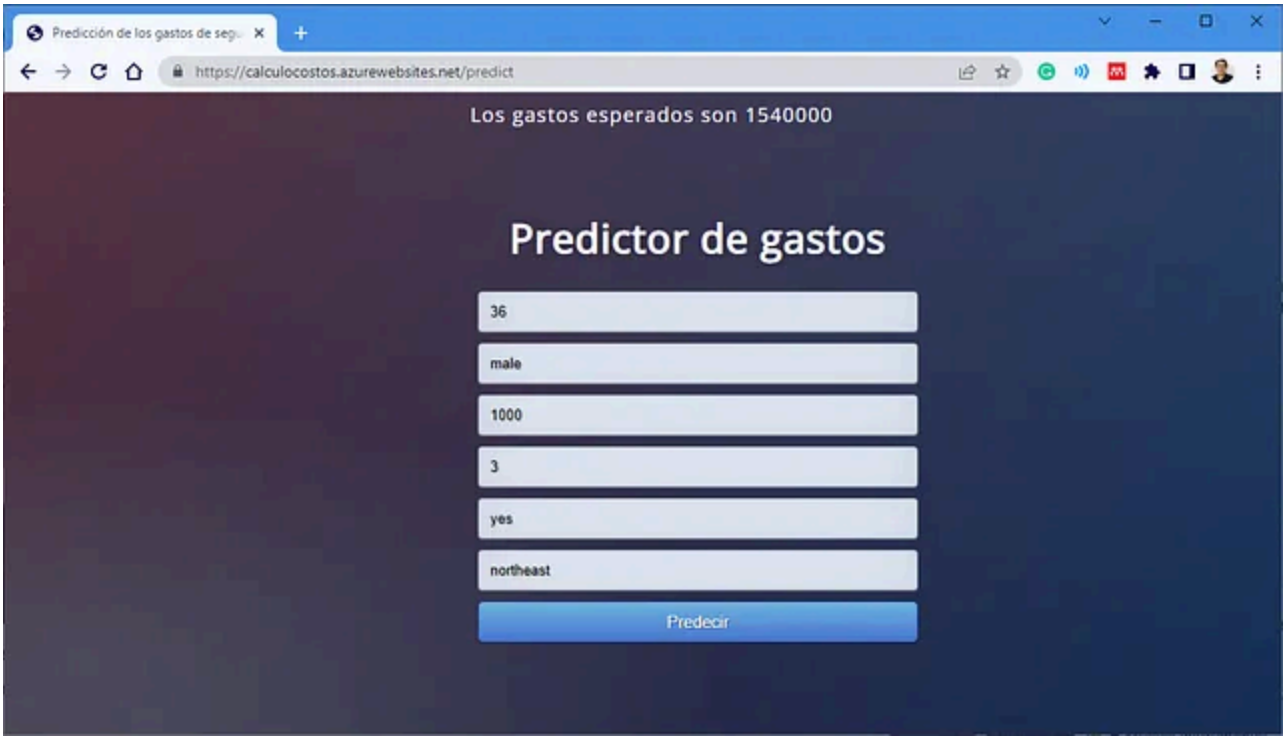
Etiqueta * : latest

Comando de inicio

Revisar y crear | < Anterior | Siguiente: Redes >



12. Test the Web Application.



References:

https://www.datacamp.com/tutorial/tutorial-machine-learning-pipelines-mlops-deployment